

UNIT 2



HCF & LCM



HCF (HIGHEST COMMON FACTOR)

The largest or greatest factor common to any two or more given natural numbers is called **Highest Common Factor (HCF)** of given numbers. It is also called **Greatest Common Divisor (GCD)**.

Highest: meaning largest number or greatest number.

Common: meaning shared between two or more numbers.

Factor: meaning an integer that a whole number can be divided by a divisor.

Real-life application:

HCF is used to find whether an event will repeat again and again.

HCF is also used to divide two or more objects into equal points or split things into similar equal sections.

Methods of HCF

There two method for finding HCF

- Prime factorization method
- Division method

1. Prime factorization

This method uses prime numbers which are the numbers that are only divisible by themselves and 1, e.g. 2, 3, 5, 7, 11, 13, 17, 19 ...

Example 1. Find HCF of 27 and 36 by using prime factorization method.

3	27
3	9
3	3
	1

2	36
2	18
3	9
3	3
	1

Prime factor of 27 = $3 \times 3 \times 3$

Prime factor of 36 = $2 \times 2 \times 3 \times 3$

HCF = $3 \times 3 = 9$

2. Division method**Example 1.** Find HCF of 66 and 88 by using division method.

$$\begin{array}{r}
 1 \\
 66 \overline{) 88} \\
 \underline{- 66} \quad 3 \\
 22 \overline{) 66} \\
 \underline{- 66} \\
 \text{X X}
 \end{array}$$

Hence HCF of 66 and 88 = 22

Example 2. Find HCF of 36, 54 and 138 by using division method.

$$\begin{array}{r}
 36 \overline{) 54} \left(1 \right. \\
 \underline{36} \\
 18 \overline{) 36} \left(2 \right. \\
 \underline{36} \\
 \text{X X}
 \end{array}$$

$$\begin{array}{r}
 18 \overline{) 138} \left(7 \right. \\
 \underline{126} \\
 12 \overline{) 18} \left(1 \right. \\
 \underline{12} \\
 6 \overline{) 12} \left(2 \right. \\
 \underline{12} \\
 \text{X}
 \end{array}$$

Hence HCF of 36, 54 and 138 = 6

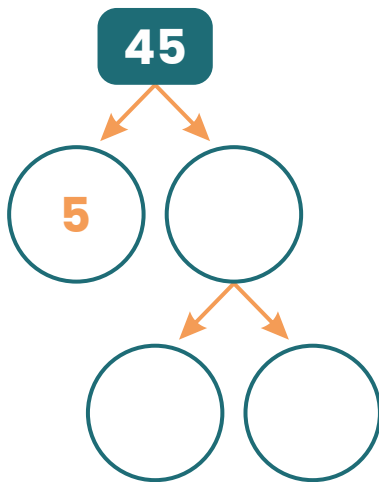
EXERCISE 2.1**Q1. Find the Highest Common Factor (HCF) or the Greatest Common Factor (GCF) of the following using the long division method:**

- 1) 48, 75 2) 78, 52, 23 3) 32, 48, 96 4) 54, 56

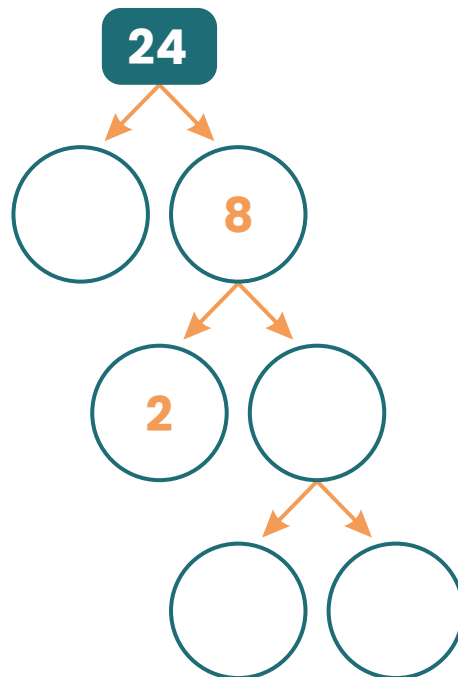
Q2. Find the HCF or GCF of the following using the prime factorization method:

- 1) 8, 64 2) 80, 45, 90 3) 12, 60, 52 4) 24, 48

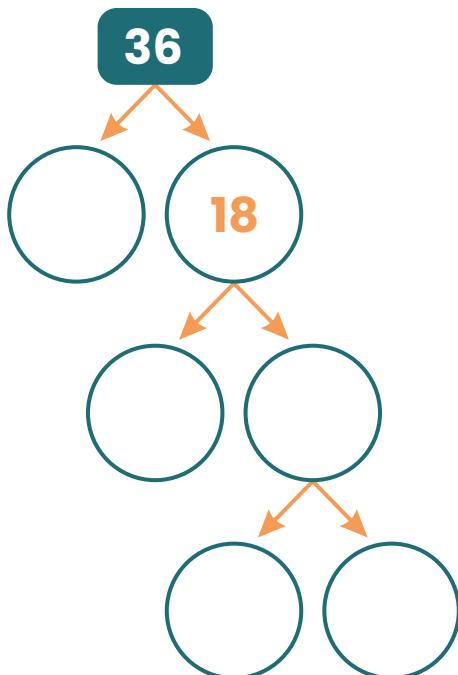
Q3. Complete the factorization to express these numbers as product of ONLY prime numbers.



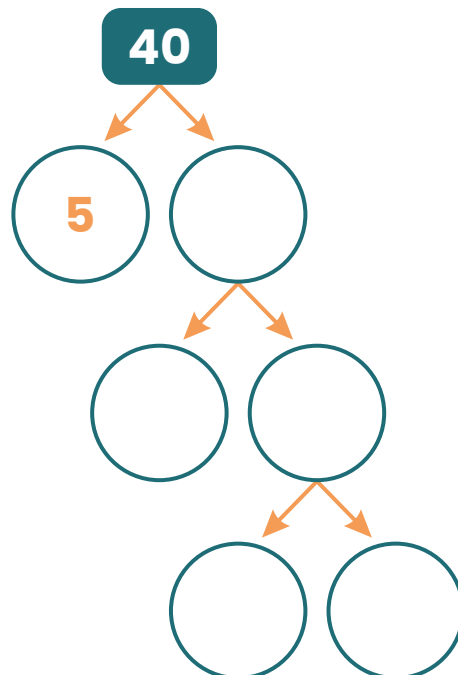
45 =



24 =

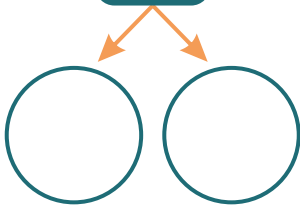


36 =

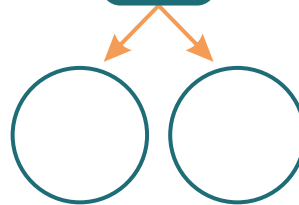


40 =

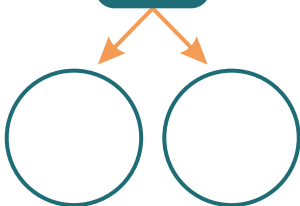
Q4. Fill in the missing numbers in these prime trees with prime factors in the circles. Then complete the prime factorization product underneath.

26

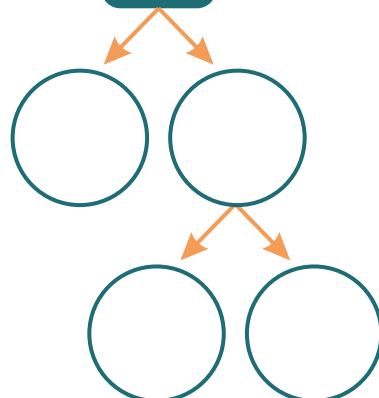
$$26 = \square \times \square$$

21

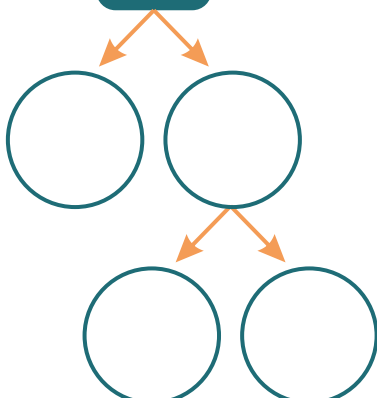
$$21 = \square \times \square$$

55

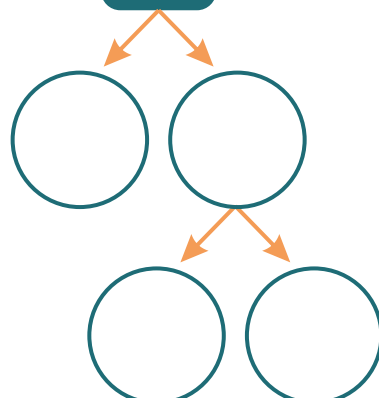
$$55 = \square \times \square$$

42

$$42 = \square \times \square \times \square$$

70

$$70 = \square \times \square \times \square$$

66

$$66 = \square \times \square \times \square$$

LCM (LEAST COMMON MULTIPLE)

The Least Common Multiple (LCM) of two or more numbers is the smallest number which can be exactly divided by each of the given numbers.

Lowest: meaning smallest number.

Common: meaning shared between two or more numbers.

Multiple: meaning the product of two numbers.

Real-life application:

LCM is used to find whether an event will happen repeatedly at the same time.

LCM is important to solve problems related to racetracks, traffic lights and load shedding etc.

Methods of LCM

There are two methods for finding LCM:

- Prime factorization method
- Division method

1. Prime factorization

Example 1. Find LCM of 44, 66 and 99 by using prime factorization method.

2	44
2	22
11	11
	1

2	66
3	33
11	11
	1

3	99
3	33
11	11
	1

Prime factorization of 44 = $2 \times 2 \times 11$

Prime factorization of 66 = $2 \times 3 \times 11$

Prime factorization of 99 = $3 \times 3 \times 11$

Product of common = 11

Product of uncommon = $2 \times 2 \times 3 \times 3 = 36$

= $11 \times 36 = 396$

2. Division method**Example 1.** Find LCM of 20 and 25 by using division method.

2	20, 25
2	10, 25
5	5, 25
5	1, 5
	1, 1

$$\text{LCM} = 2 \times 2 \times 5 \times 5 = 100$$

EXERCISE 2.2**Q1. Find Lowest common multiple (LCM) by prime factorization method.****1)** 16, 24, 40**2)** 84, 144**3)** 72, 98, 120**Q2. Find LCM of the following by division method.****1)** 70, 110, 150**2)** 25, 30, 150**3)** 36, 60, 120**Q3. Find LCM for each of the following pairs of numbers by division method.****1)** 18 and 27**2)** 48 and 60

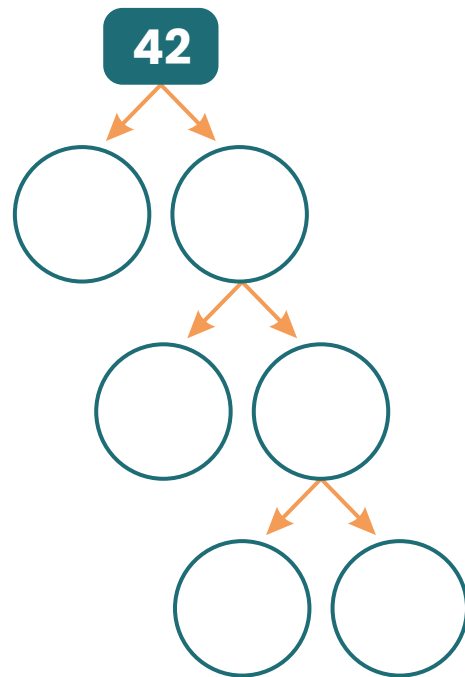
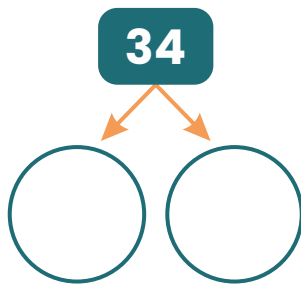
	18, 27

LCM = _____

	48, 60

LCM = _____

Q4. Find prime factors of each of the following.



Q5. Solve the following.

• Find the L.C.M. of the following:

1) 54, 60 and 90

2) 34, 21 and 14

3) 34, 119 and 196

4) 20, 32 and 40

5) 24 and 30

6) 15 and 20

7) 24 and 3

8) 48 and 72

9) 30 and 45

10) 18 and 54